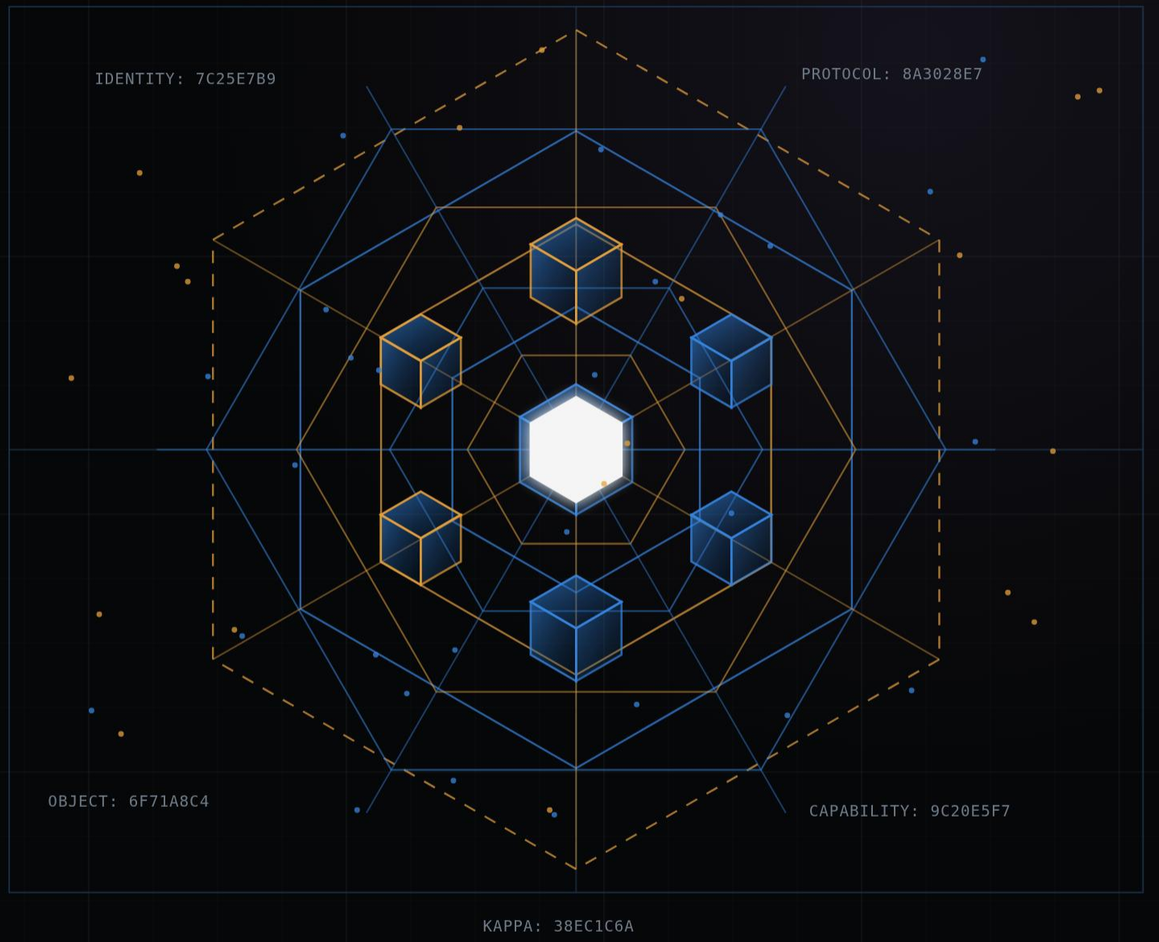
The Object Internet

Agents don't need more pages.
They need objects they can **trust**.



NANDA × UOR

The Agentic Web — A Reframe

Today's agentic internet is **file-oriented**. The object internet is **object-oriented**.

1

Files → Objects

2

Scraped Text → DataFacts

3

URLs → Structural Addresses

4

API Calls → Verifiable Exchanges

Don't give agents more websites to browse — give them objects they can identify, verify, route, and act on.

`uor-addr` is the content addressing standard

It turns content into a stable address

`[V1] {"a":1} vs {"a":1.0}`	→	`[PASS]`	Identical Address (Key / type invariance)
`[V2] String NFC vs NFD`	→	`[PASS]`	Identical Address (Unicode invariance)
`[V3] {"a":1} vs {"a":2}`	→	`[PASS]`	Different Address (Tamper sensitivity)
`[V4] Empty structural content`	→	`[REJECT]`	CanonicalizationError

Where Byte-Level Integrity Breaks

Agents reserialize the same content — key order, int vs. float, unicode form. Byte identity forks while the fact is unchanged.

Provenance and quarantine flags are silently severed.

Byte Checksum

```
A → e0ebb4ff…
B → fe0b8b53…
```

Two addresses. History forks.

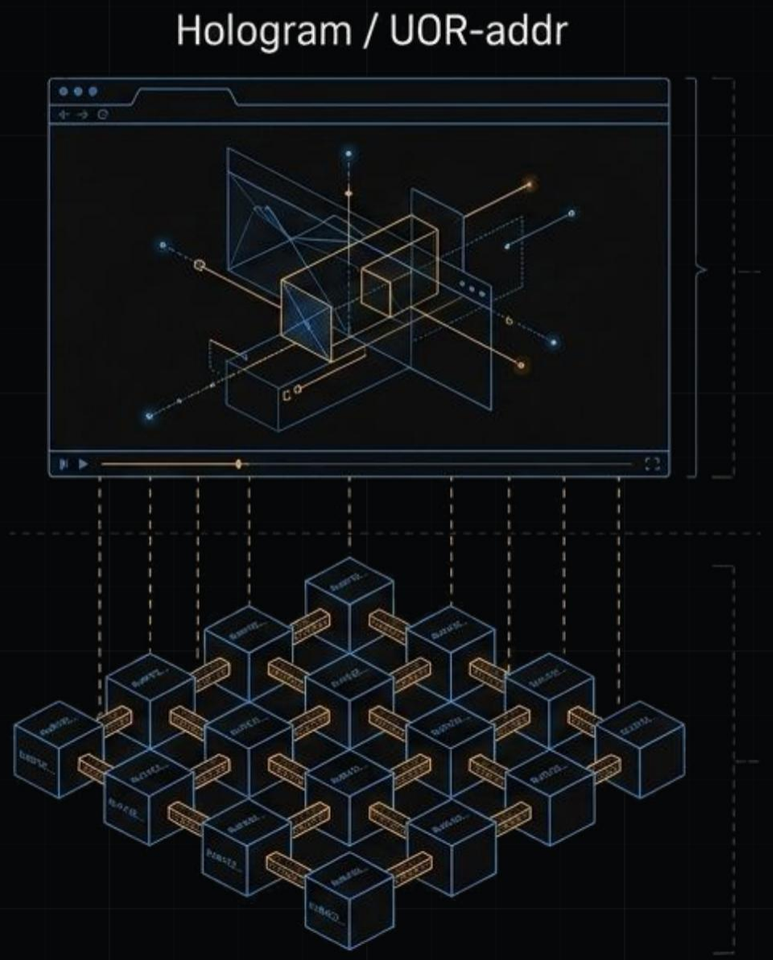
Structural Address

```
A → 9c038679… = B → 9c038679…
```

One address. History holds.

The viewer owns the meaning

It asks for a verified object. The registry finds the bytes.



Hologram + UOR protocol

The viewer chooses or creates the protocol above kappa-registry, identifies it with UOR-ADDR, and rebuilds the exact meaning the agent needs.

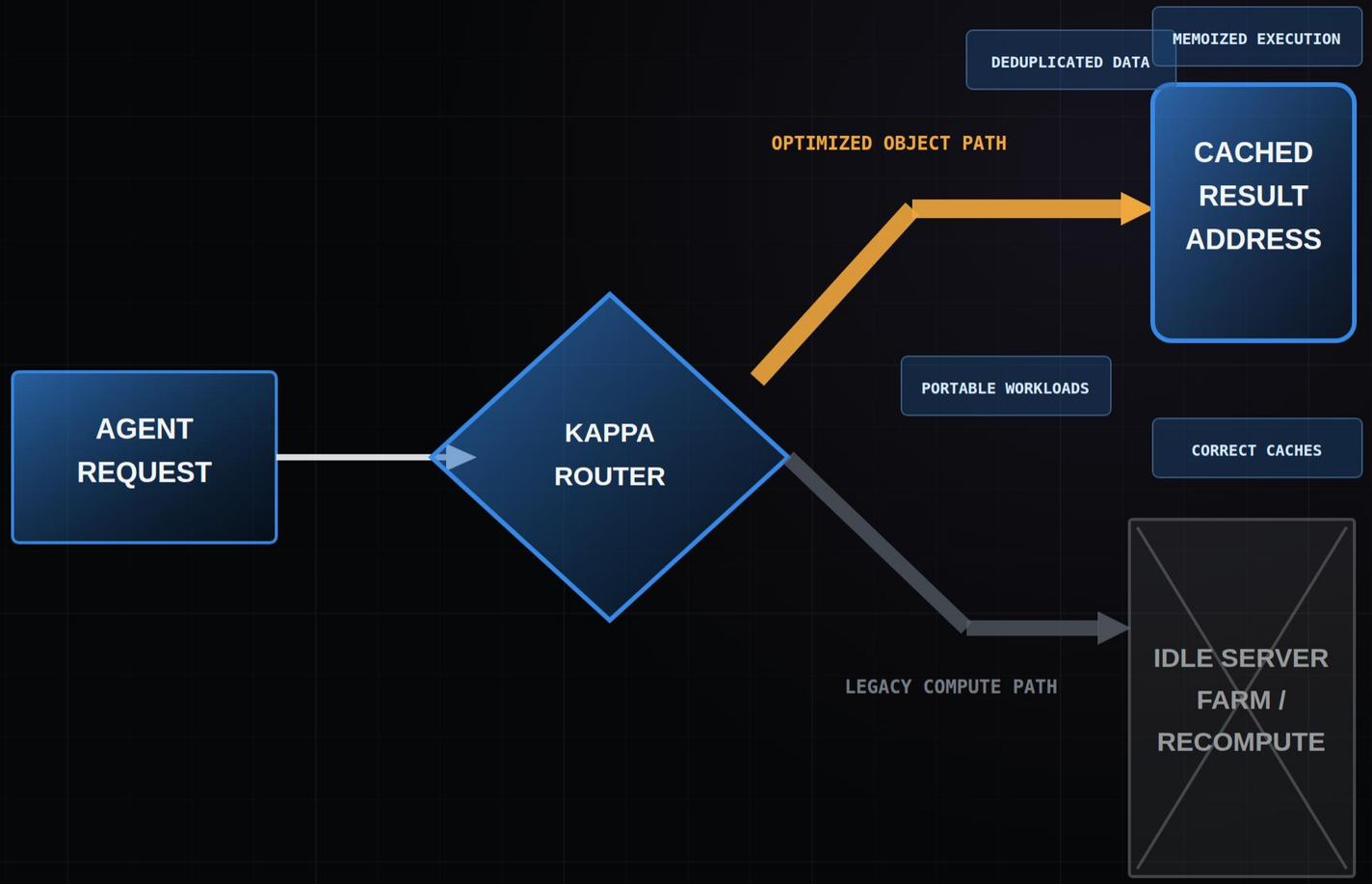
kappa-registry / OS

Resolves, verifies, traverses, deduplicates, and retrieves the object—without exposing buckets, paths, shards, or peer locations.

An address becomes a routing decision

Resolve what is already verified.
Compute only what is missing.

Strongest for repeated, cacheable, portable, verifiable work.



Compute starts with what already exists

Old question: Which server should run this?

New question: What has already been verified—and what is missing?

`[Input Address] + [Function Address] + [Environment Address] = [Result Address]`

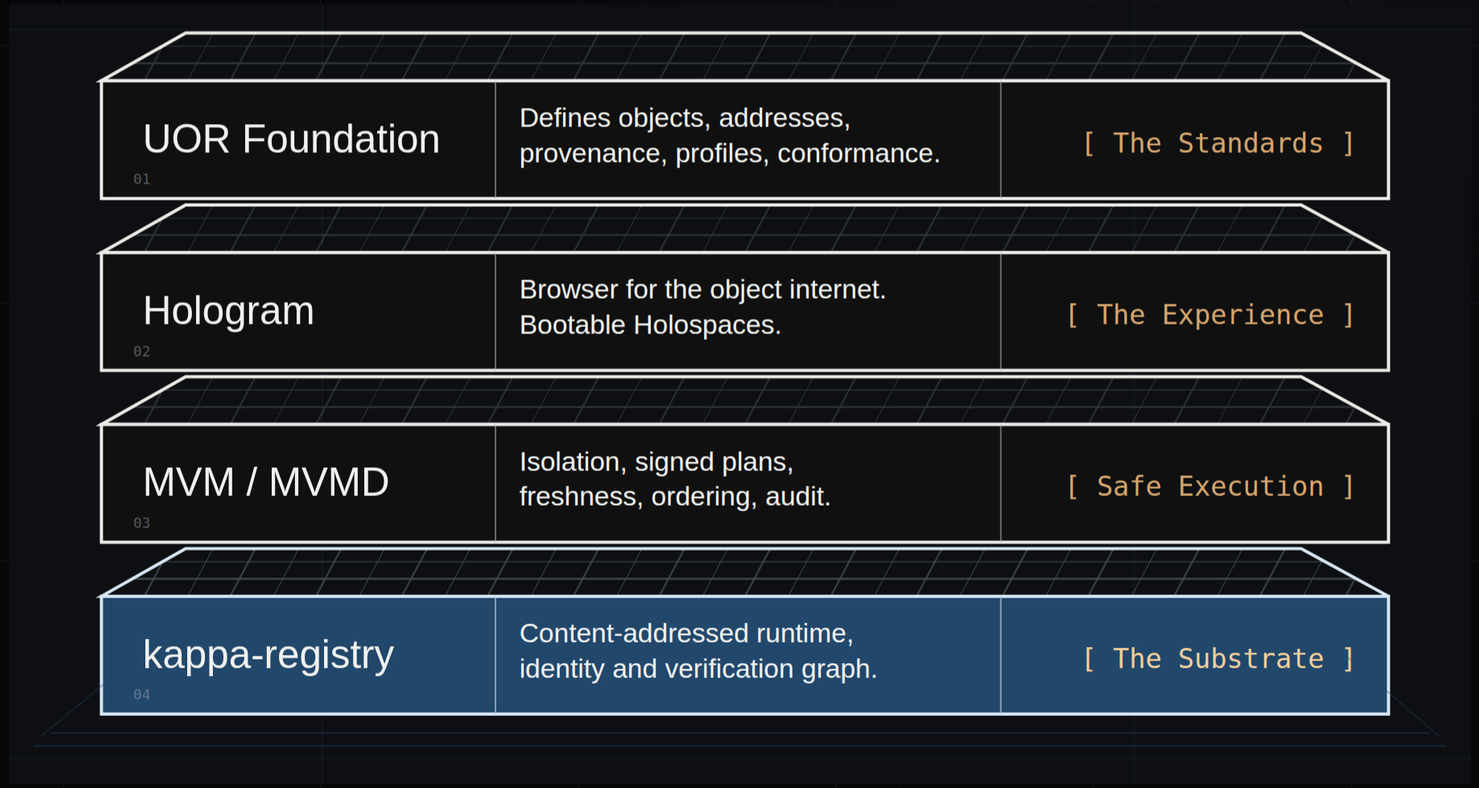
Once *sameness* is recognized through object identity, systems stop copying, rebuilding, recomputing, and re-verifying from scratch. Compute moves by resolving closures—not copying whole machines.

Agents do not need more websites to browse. They need objects they can: identify, verify, route, transform, and act on

Files → Objects

Endpoints → Proof

Cloud jobs → Verified reuse



The action status

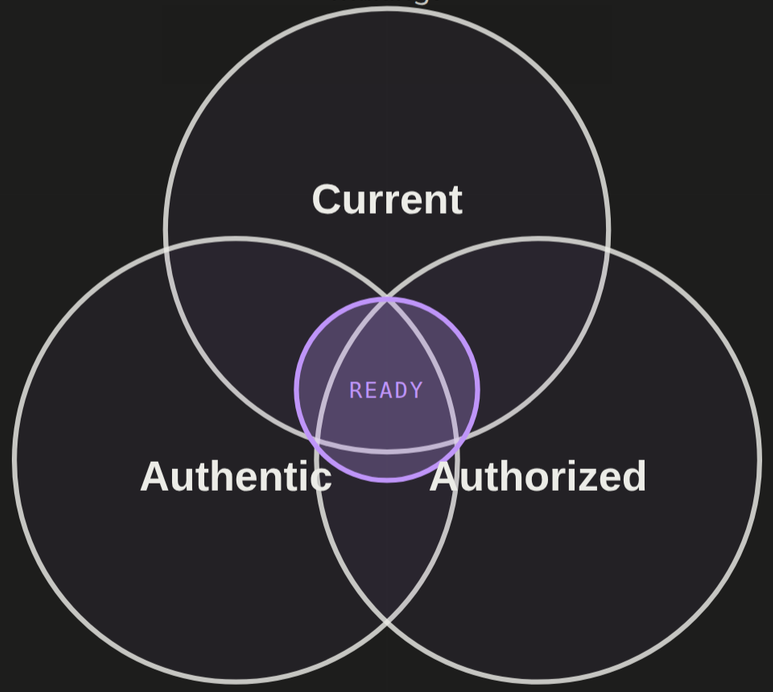
Data Facts asks one defining question: Is the agent’s data ready for decision-making?

Current

Is the data fresh enough for decision-making?

Authentic

Is the identity and integrity of the source intact?



Authorized

Does this specific agent have permission to access it?

The Triad: From Evidence to Binding

Data Facts asks whether evidence is ready to use. UOR-NANDA makes each answer verifiable and executable.

Data Facts asks	UOR-NANDA provides
01 **Is the data current?**	**Signed timestamp + append-only log** ◌
02 **Is the data authentic?**	**Structural content address** #
03 **Is the data authorized?**	**Capability chain + revocation checks** ↗

Evidence becomes binding. The original field stays intact; the proof travels with the object.

UOR-NANDA Closes the Loop



Sever implements the triad live — blind-run tested by OpenClaw agents. Current = signed timestamp + log. Authentic = structural address. Authorized = capability chain + revocation checks.

The accountability loop, running live



▶ Run the loop | Step → | Reset

in-browser · WebCrypto · no backend

The verifier recomputes the whole chain in your browser: permission real, still valid, output matches the address. The escrow releases.

POST /verify

✓ ALLOW · escrow released

copy


```
{
  "decision": "ALLOW",
  "xrpl_tx_hash": "63E35A5852408699342102FFD08F371DC443973FBB97001AA32728CF69AEBD03",
  "explorer_url": "https://testnet.xrpl.org/transactions/63E35A5852408699342102FFD08F371DC443973FBB97001AA32728CF69AEBD03"
}
```


↳ recorded on the public testnet: testnet.xrpl.org/transactions/63E35A5852408699…

What the Object Internet Unlocks

Memory

An object with identity, provenance, freshness, permissions — not “remembered some text”

Tool Use

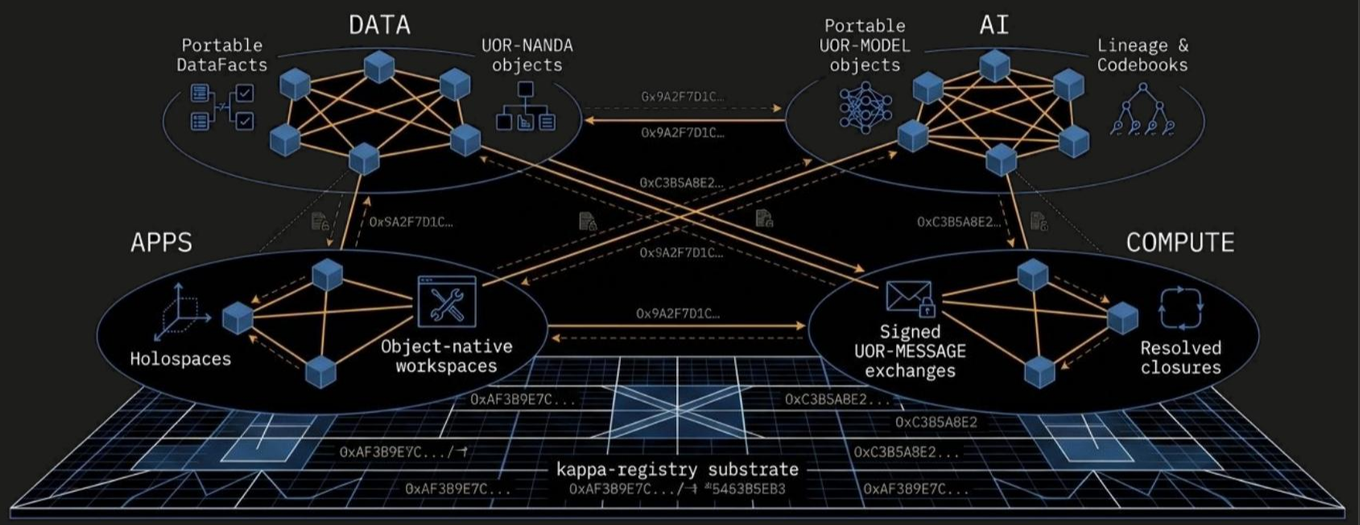
Acted on a verified object under a capability — not “called an API”

Multi-Agent Comms

Signed, verifiable messages and codebooks — not “exchanged messages”

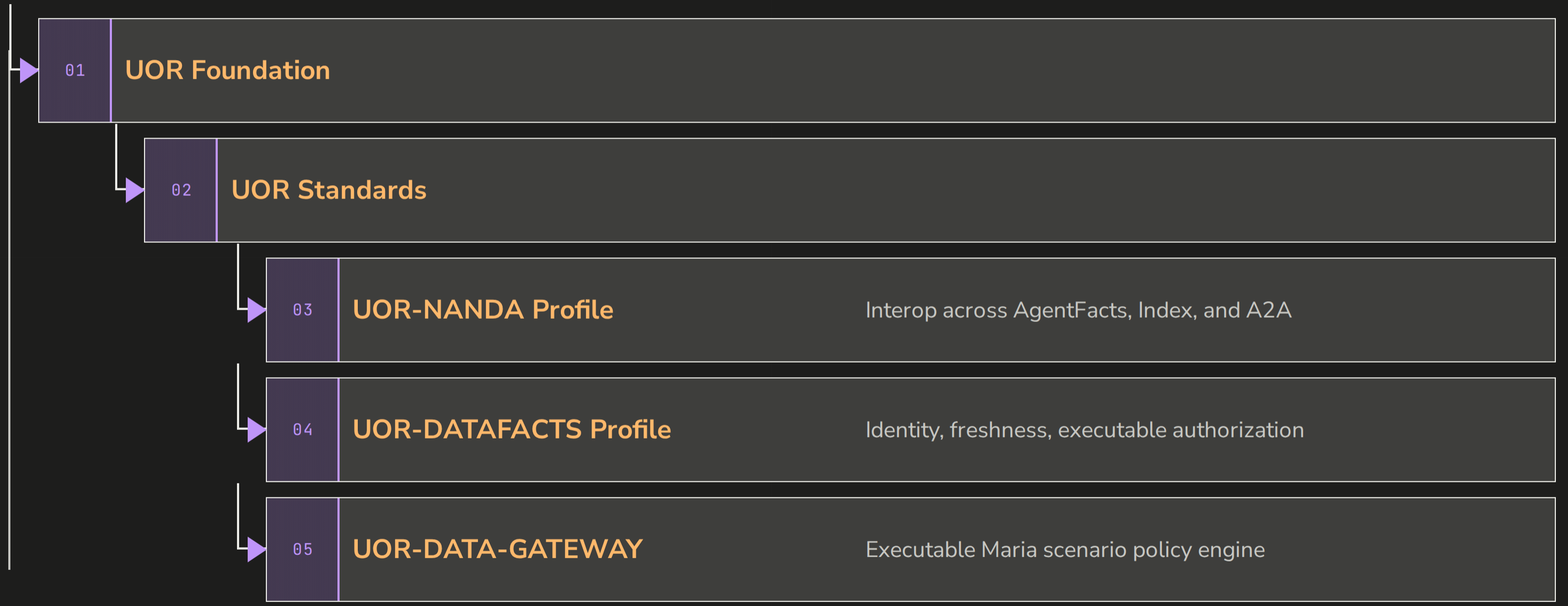
Compute

Resolve the closure, compute only what’s missing — not “run it again somewhere”



Operating Framework Placement

Standardizing the external ecosystem profiles for immediate interoperability and governance.



The placement matters. UOR provides the substrate, profiles make it interoperable, and the gateway makes governance executable.

Thank you

Ari Lerner and the Hologram and UOR teams

```
import { kappa } from "@uor-foundation/uor-addr";

const label = kappa.jsonAddress(
  new TextEncoder().encode('{"msg":"Thank you"}')
);
console.log(label);
// sha256:8839aafda444d2c36d1b3f27d84ef287f6261975b1b6ddc9972054854b6
```

Project NANDA
The Internet of AI Agents

